The Exact Local Coalition Game of Sparse Mixture-of-Experts

Exhaustive Counterfactual Measurement Reveals Non-Additive Expert Damage

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Abstract

Sparse mixture-of-experts (MoE) models activate only a small expert coalition for each token, yet adaptive execution and pruning methods often compress expert value into independent scalar scores or approximate coalition statistics. We exploit the sparsity itself. Conditional on the naturally selected top-k coalition, we exhaustively replay every execution subset through the frozen downstream model and recover the complete state-conditioned local damage function. For gpt-oss-120b, where $k = 4$, the local game has only $2^4 = 16$ subsets even though each MoE layer contains 128 experts. Across five intervention layers, two corpora, eight independent captures, and 801 unique states, we measure every local subset using direct downstream final-logit KL divergence. The first-order singleton approximation correlates only 0.6425 with exact higher-order subset damage, has 43.79% median relative absolute error and 321.88% 95th-percentile relative error, and the measured damage function violates monotonicity on 20.91% of subset-lattice edges. Contextual marginal damage for the fourth-ranked expert changes sign in 21.55% of contextual marginal evaluations. At identical expert count, the best three-expert subset reduces mean KL by 3.62× relative to the fixed rank-based three-expert policy; the corresponding two-expert gap is 3.59×. We formalize the local game with Boolean-lattice Möbius inversion, prove that additive approximation error is exactly higher-order interaction mass, relate exact Shapley attribution to Möbius coefficients, and prove by counterexample that even exact Shapley rankings are not sufficient to recover a fixed-cardinality subset optimum. The finite four-player identities and the Shapley counterexample are machine-checked in Lean 4.32.2. Our results do not provide a deployable subset predictor; rather, they identify the exact decision object that such a predictor must approximate.

1 Introduction

Sparse MoE architectures separate total model capacity from per-token computation by routing each token through only a small subset of experts [Shazeer et al., 2017]. The same sparsity that makes MoEs computationally attractive creates a difficult deployment question: once a router has selected a small active set, which of those expert computations are actually necessary for preserving the downstream model distribution?

A natural simplification is to assign each expert a scalar importance and make execution or pruning decisions from those scores. Recent work has already recognized that this can miss coalition effects. Shapley-MoE formulates expert pruning as a cooperative game and approximates global Shapley values with Monte Carlo sampling because exhaustive evaluation over the full expert pool is intractable [Huang et al., 2025]. SHAPE explicitly models routed expert coalitions, but deliberately avoids additional model evaluations and instead constructs a Shapley-style proxy from routing frequencies and co-occurrence statistics [Zhang, 2026]. Counterfactual routing analysis has also compared native routes against sampled equal-compute alternatives and demonstrated that lower-loss alternatives can exist in a frozen MoE [Yoon et al., 2026]. Diversity-aware routing methods such as MP-MoE likewise formulate expert routing as a subset-selection problem [Kang et al., 2026].

Complementary recent work studies expert collaboration through co-activation structure [Tang et al., 2026] and allocates constrained expert activations across layers and tokens [Liu et al., 2026].

We study a different object: the *complete local game after routing*. Let the natural router select an active set $\mathcal{A}_h$ of k experts at hidden state h. Rather than valuing all experts in the layer or sampling alternative routes, we hold the naturally selected expert identities fixed and evaluate every subset of $\mathcal{A}_h$. The global expert game may be exponentially impossible, but the local game contains only 2^k actions. For gpt-oss-120b, each MoE layer has 128 experts but activates four per token [OpenAI, 2025]; therefore the local game has 16 actions.

This conditioning step changes the tractability of exact coalition measurement. It does not solve global routing, and it does not evaluate experts outside the natural active set. It instead asks the execution-time question that remains *after* the router has reduced a 128-expert layer to four candidates:

> Given the experts already selected for this state, what is the exact downstream damage of executing each possible survivor subset?

We answer that question by intervention on the native gpt-oss-120b MXFP4 checkpoint. For each sampled state, we preserve the natural top-four expert identities, renormalize the original router weights over each nonempty survivor subset, replay the remaining transformer layers, and measure $\mathrm{KL}(p_{\mathrm{full}} \| p_{\mathrm{subset}})$ at the final logits. Executing no routed experts is treated separately as an explicit zero-MoE-correction runtime action.

Our main contributions are:

1. We define and exhaustively measure the *state-conditioned local omission game* over the naturally active top-four coalition, yielding the full 16-value downstream damage table for 801 unique gpt-oss-120b states.

2. We show that the measured game is strongly non-additive and frequently non-monotone. Independent singleton damages are not reliable surrogates for joint omission damage.

3. We quantify a large same-compute oracle gap: choosing the best subset at a fixed expert count reduces mean KL by approximately 3.6× relative to fixed rank-based truncation for both two- and three-expert execution.

4. We formalize the finite game with Möbius interactions [Rota, 1964]. We prove that first-order approximation error is exactly the sum of higher-order interaction terms and connect the complete game to exact Shapley attribution [Shapley, 1953].

5. We prove that even exact Shapley rankings are not, in general, sufficient statistics for cardinality-constrained subset optimization. The algebraic identities and a four-player counterexample are checked in Lean 4.32.2.

The practical conclusion is intentionally limited. Exact local enumeration establishes the decision target and an oracle opportunity; our first cheap held-out predictor does not generalize well enough for deployment. We therefore separate *measurement of the decision object* from *prediction of that object at runtime*.

2 The State-Conditioned Local Game

2.1 Natural active coalition

Consider one MoE layer at a captured hidden state h. Let the router naturally select

$$\mathcal{A}_h = \{1, \dots, k\}, \qquad k = 4,$$

with normalized top-k router weights $g_i > 0$ satisfying $\sum_{i \in \mathcal{A}_h} g_i = 1$. Let $E_i(h)$ denote the output of selected expert i on the captured state.

For a nonempty executed survivor set $S \subseteq \mathcal{A}_h$, the intervention retains the natural selected identities and renormalizes the original router weights:

$$\widetilde{g}_i^{(S)} = \frac{g_i}{\sum_{j \in S} g_j}, \qquad i \in S. \tag{1}$$

The routed expert correction is

$$m_S(h) = \sum_{i \in S} \widetilde{g}_i^{(S)} E_i(h). \tag{2}$$

For the empty survivor set we define a separate extended runtime action

$$m_\varnothing(h) = 0, \tag{3}$$

which means "execute no routed expert correction." Equation (3) is not survivor renormalization and must be distinguished from the 15 nonempty execution actions.

Let p_h^S be the final next-token distribution obtained after inserting $m_S(h)$ at the intervention layer and replaying every remaining decoder layer through the frozen checkpoint. The full natural path is $p_h^{\mathcal{A}_h}$.

2.2 Omission damage

The implementation and all reported interaction statistics are indexed by an *omission set*. For $O \subseteq \mathcal{A}_h$, define

$$D_h(O) = \mathrm{KL}\left(p_h^{\mathcal{A}_h} \middle\| p_h^{\mathcal{A}_h \setminus O}\right). \tag{4}$$

Thus $D_h(\varnothing) = 0$. For $O \neq \mathcal{A}_h$, the survivor set is nonempty and Eq. (1) defines the intervention exactly. For $O = \mathcal{A}_h$, the value uses the explicit zero-correction action in Eq. (3).

This distinction matters for interaction interpretation. All singleton, pair, and triple omission values are measured under proper survivor renormalization. Any four-way Möbius term necessarily touches $D_h(\mathcal{A}_h)$ and therefore depends on the explicitly chosen zero-correction extension.

2.3 Runtime objective

Let $C_h(S)$ be the state- and scheduler-dependent cost of executing survivor set S, including effects such as cache residency, transfers, shared staging, and overlap. Given a tradeoff $\lambda \geq 0$, the exact local runtime problem is

$$S_h^*(\lambda) = \arg\min_{S \subseteq \mathcal{A}_h} \left[D_h(\mathcal{A}_h \setminus S) + \lambda C_h(S) \right]. \tag{5}$$

For $k = 4$, Eq. (5) contains 16 candidates. Once the damage table and costs are known, enumeration is exact; no modularity or independent-expert assumption is required.

3 Möbius Interactions and Additive Error

A set function on the Boolean lattice has a unique Möbius representation [Rota, 1964]. For the measured omission-damage function, define

$$\mu_h(T) = \sum_{U \subseteq T} (-1)^{|T|-|U|} D_h(U). \tag{6}$$

The coefficient $\mu_h(T)$ is the interaction associated with omission set T. For example,

$$\mu_h(\{i\}) = D_h(\{i\}) - D_h(\varnothing), \tag{7}$$

$$\mu_h(\{i,j\}) = D_h(\{i,j\}) - D_h(\{i\}) - D_h(\{j\}) + D_h(\varnothing). \tag{8}$$

Because $D_h(\varnothing) = 0$, singleton coefficients equal singleton omission damages.

Proposition 1 (Möbius reconstruction)**.** *For every $O \subseteq \mathcal{A}_h$,*

$$D_h(O) = \sum_{T \subseteq O} \mu_h(T). \tag{9}$$

Proof. Substitute Eq. (6) into the right side:

$$\sum_{T \subseteq O} \mu_h(T) = \sum_{T \subseteq O} \sum_{U \subseteq T} (-1)^{|T|-|U|} D_h(U).$$

Swap the finite sums and collect the coefficient of each $D_h(U)$:

$$\sum_{U \subseteq O} D_h(U) \sum_{U \subseteq T \subseteq O} (-1)^{|T|-|U|}.$$

For fixed U, subsets T between U and O correspond to subsets of $O \setminus U$, so the inner sum is

$$\sum_{r=0}^{|O \setminus U|} \binom{|O \setminus U|}{r} (-1)^r = (1-1)^{|O \setminus U|}.$$

It is one only when $U = O$ and zero otherwise. The remaining term is $D_h(O)$. □

The first-order or modular approximation retains only singleton coefficients:

$$\widehat{D}_h^{(1)}(O) = \mu_h(\varnothing) + \sum_{i \in O} \mu_h(\{i\}). \tag{10}$$

Here $\mu_h(\varnothing) = D_h(\varnothing) = 0$, so Eq. (10) is exactly the sum of singleton omission damages used in our empirical additive baseline.

Proposition 2 (Additive error is higher-order interaction)**.** *For every omission set O,*

$$D_h(O) - \widehat{D}_h^{(1)}(O) = \sum_{\substack{T \subseteq O \\ |T| \geq 2}} \mu_h(T). \tag{11}$$

Proof. Apply Eq. (9) and partition the reconstruction sum into terms of order zero, order one, and order at least two. Subtracting Eq. (10) cancels the first two groups exactly. □

Equation (11) is important conceptually: disagreement between singleton-additive prediction and measured joint damage is not an ambiguous residual. It is exactly the aggregate higher-order interaction contribution on that subset.

4 Shapley Attribution Is Not Subset Optimization

Shapley values provide a principled scalar attribution of a complete cooperative game [Shapley, 1953]. This is relevant to MoE pruning because Shapley-MoE explicitly estimates global expert Shapley values [Huang et al., 2025], while SHAPE uses a routing-trace-based Shapley-style approximation [Zhang, 2026]. Our exact local measurement makes a complementary distinction possible: even if the Shapley values of the local game were known exactly, a scalar Shapley ranking does not generally encode the cardinality-constrained subset optimum.

To state the standard result, let $v : 2^N \to \mathbb{R}$ be an arbitrary cooperative game with $v(\varnothing) = 0$ and Möbius coefficients μ_v. The Shapley value of player i is

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(n - |S| - 1)!}{n!} \left[v(S \cup \{i\}) - v(S) \right]. \tag{12}$$

Proposition 3 (Shapley–Möbius identity)**.** *For every player i,*

$$\phi_i(v) = \sum_{\substack{T \subseteq N \\ i \in T}} \frac{\mu_v(T)}{|T|}. \tag{13}$$

Proof. Interpret Eq. (12) as the expected marginal contribution of player i under a uniformly random permutation of the players. Expand v into Möbius interactions. An interaction term $\mu_v(T)$ with $i \notin T$ cancels from the marginal. If $i \in T$, the interaction appears in the marginal exactly when every other member of T precedes i in the random permutation, i.e. when i is the last member of T to arrive. By symmetry, each of the $|T|$ members is equally likely to be last, so the probability is $1/|T|$. Taking expectations gives Eq. (13). □

Equation (13) shows what Shapley attribution does to interactions: each higher-order Möbius term is distributed evenly among its participating players. That is a valid attribution rule, but it compresses the full 2^n table into n scalars.

Proposition 4 (Exact Shapley ranking need not recover the best pair)**.** *There exists a four-player game whose exact Shapley ranking is*

$$1 > 2 > 3 > 4,$$

but whose highest-value two-player coalition is $\{2, 3\}$ rather than the top-two Shapley coalition $\{1, 2\}$.

Proof. Define the nonzero Möbius coefficients

$$\mu_v(\{1\}) = 100, \qquad \mu_v(\{2\}) = 2, \qquad \mu_v(\{2, 3\}) = 180,$$

and set all other coefficients to zero. Möbius reconstruction gives

$$v(\{1, 2\}) = 102, \qquad v(\{2, 3\}) = 182,$$

so $\{2, 3\}$ is the best two-player coalition. From Eq. (13),

$$\phi_1 = 100, \qquad \phi_2 = 2 + \frac{180}{2} = 92, \qquad \phi_3 = \frac{180}{2} = 90, \qquad \phi_4 = 0.$$

Thus the exact Shapley ranking selects players 1 and 2, whose coalition value is 102, while the optimal pair $\{2, 3\}$ has value 182. □

Corollary 1. *For arbitrary non-additive local MoE games, exact per-expert Shapley values are not sufficient statistics for the exact fixed-cardinality execution optimum.*

This proposition does not say that Shapley values are poor expert attributions, nor does it evaluate Shapley-MoE or SHAPE empirically. It establishes a narrower mathematical point: attribution and combinatorial optimization are different objects. A complete local game retains interaction placement that any n-scalar attribution necessarily compresses.

For direct comparison to the measured omission game, one may define an execution-value game by an affine re-indexing such as

$$V_h(S) = D_h(\mathcal{A}_h) - D_h(\mathcal{A}_h \setminus S), \tag{14}$$

which satisfies $V_h(\varnothing) = 0$ and ranks execution subsets in exactly the reverse order of damage. Because V_h includes the empty-execution baseline through $D_h(\mathcal{A}_h)$, its full-order attribution inherits the same explicit zero-correction convention discussed in Sec. 2.

5 Experimental Method

5.1 Model and intervention layers

We use the official gpt-oss-120b checkpoint in its deployed MXFP4 representation. OpenAI reports that gpt-oss-120b has 36 transformer layers, 128 experts per MoE layer, and four active experts per token [OpenAI, 2025]. The experiment intervenes at layers 16, 17, 19, 32, and 33.

We evaluate two distinct token corpora using eight independent captures. Across the captures there are 960 raw observations. Deduplication by intervention layer, corpus, and token position yields 801 unique natural states.

5.2 Exhaustive local replay

For each natural state:

1. Recompute the native router logits and record the natural top-four expert IDs and normalized weights.
2. Hold those four expert identities fixed.
3. For each of the 15 nonempty survivor masks, renormalize the native top-four weights according to Eq. (1) and execute the native expert mixture.
4. Define the sixteenth action, the empty survivor set, as zero routed-expert correction.
5. Insert each counterfactual correction at the captured intervention state and replay every remaining decoder layer using the frozen native model.
6. Compute the final next-token logits for the same token position and evaluate Eq. (4) against the full top-four path.

This yields 11,214 proper survivor-renormalization omission counterfactuals (801×14), plus 801 explicit zero-correction counterfactuals, for 12,015 nontrivial counterfactual executions in total. Together with $D_h(\varnothing) = 0$, every state has a complete 16-entry local damage table.

The intervention does not alter expert parameters or router parameters, does not train a proxy model, and does not substitute experts from outside the natural top-four coalition. It measures the frozen checkpoint's downstream response to local execution choices.

Table 1: First-order singleton approximation versus exact subset damage.

Statistic	Result
Correlation with exact subset KL	0.6425
Median relative absolute error	43.79%
95th-percentile relative absolute error	321.88%
Mean absolute error	0.05016 nat
Underestimation fraction	50.71%
Damage-monotonicity violation fraction	20.91%

5.3 Evaluation quantities

We report four families of quantities.

Approximation quality. We compare measured $D_h(O)$ against the first-order reconstruction $\widehat{D}_h^{(1)}(O)$ for subsets of order at least two.

Interaction structure. We compute Eq. (6), reconstruct the 16-value table, inspect interaction signs, measure subset-lattice monotonicity, and compare contextual marginal damage with singleton damage.

Fixed-cardinality oracle gap. For exactly three executed experts, the fixed rank-based policy always omits rank four, whereas the oracle selects the least-damaging one-expert omission. For exactly two executed experts, the fixed rank-based policy executes ranks one and two, whereas the oracle selects the best of all six two-expert survivor sets. Both policies therefore use identical expert count.

Runtime decision gap. We combine the damage table with measured uniform-k runtime costs and sweep a cost penalty λ. The diagnostic sweep is restricted to $|S| \in \{2, 3, 4\}$, the execution counts for which measured uniform-k costs are available. This tests how often an additive approximation would choose a different action from exact finite-set enumeration. The cost model is diagnostic; the scientific claim is the disagreement induced by damage non-additivity, not a universal hardware cost law.

6 Results

6.1 The singleton-additive model is inaccurate

Table 1 compares the first-order reconstruction with exact higher-order subset KL.

The median error alone rules out treating singleton damage as an accurate decomposition of the measured local game. The tail is substantially worse: the 95th-percentile relative error exceeds three times the exact damage. Under- and over-estimation are both common, so the failure is not a simple calibration bias.

The 20.91% monotonicity-violation rate is also important. Here monotonicity means that omitting an additional expert should not reduce $\mathrm{KL}(p_{\mathrm{full}} \| p_{\mathrm{subset}})$. The observed violations show that local damage is not generally monotone on the omission lattice under survivor renormalization

Table 2: Fixed rank-based execution versus exact same-cardinality oracle. Lower KL is better.

Executed experts	Fixed is oracle	Oracle mean KL	Fixed mean KL	Ratio
3	44.94%	0.00578	0.02091	3.62×
2	36.70%	0.01411	0.05063	3.59×

and downstream replay. This is compatible with destructive interference or compensating effects among executed experts; it is not evidence that "fewer experts are globally better."

6.2 Higher-order interactions have both signs

The exact Möbius transform reconstructs every measured 16-value state table with zero numerical reconstruction error. Pair interactions are approximately balanced between positive and negative signs. Triple terms are positive in 65.29% of measured cases. The reported four-way term is negative in 87.02% of states, but this statistic must be interpreted with the caveat from Sec. 2: it includes the explicitly defined zero-correction action and is therefore extension-dependent.

Context dependence is visible even for one selected rank. Let expert four denote the router's fourth-ranked member of the natural top-four set. Its contextual marginal damage across the possible omission contexts correlates only 0.4359 with its singleton omission damage and changes sign in 21.55% of contextual marginal evaluations. A scalar singleton estimate can therefore assign the wrong direction to the expert's marginal effect in a substantial fraction of local contexts.

6.3 The same-compute oracle gap is large

Table 2 compares fixed rank truncation with the best subset at the exact same executed expert count.

For three-expert execution, always dropping router rank four is exactly optimal in fewer than half of states. At the same expert count, exact subset choice lowers mean KL by a factor of 3.62. The two-expert result is similar: executing router ranks one and two is optimal in only 36.70% of states, and the exact pair lowers mean KL by 3.59×.

These ratios are oracle measurements, not achieved runtime speedups. Their purpose is to measure the value of the decision target while holding compute cardinality fixed.

6.4 Exact subset choice enlarges the fidelity-feasible region

At a fidelity budget of 0.003 nat, allowing the best three-expert subset makes an additional 14.86% of states feasible relative to always dropping rank four. For two experts, exact pair selection recovers an additional 9.24% of states. At a 0.01-nat budget, the corresponding recoverable fractions are 9.99% and 13.86%.

When the singleton-additive approximation is used to make the subset decision, its nominal fidelity constraint can fail on the true downstream game. The actual violation rate is 11.49% at a 0.003-nat threshold and 14.86% at 0.01 nat. Across the measured cost-penalty sweep, the additive policy chooses a different action from exact enumeration in as many as 57.80% of states.

6.5 The first cheap predictor fails the deployment gate

The exact local game is useful only operationally if it can eventually be predicted more cheaply than exhaustive downstream replay. As a first test, we trained an ExtraTrees baseline from pre-execution

features: layer, selected expert identities, router weights and margins, router entropy, and a fixed random hidden-state sketch. Training on one corpus and testing on the other yields subset-damage correlations of only 0.1653 and 0.0679 in the two directions. One held-out 0.003-nat configuration has a 54.70% true violation rate.

We therefore do not claim a deployable adaptive execution method. The failed predictor is a guardrail: the oracle gap demonstrates that the target is valuable, while the held-out failure demonstrates that simple pre-execution features are not enough to realize it safely.

7 Relation to Prior Coalition-Aware MoE Work

Our novelty claim is deliberately narrower than "experts interact." Existing work already makes that point in several forms.

Shapley-MoE. Shapley-MoE defines expert pruning through cooperative-game Shapley values and explicitly avoids exhaustive global subset evaluation by using Monte Carlo estimation, early truncation, and router-guided importance sampling [Huang et al., 2025]. The object is a global pruning valuation over a large expert pool. Our experiment instead conditions on the four experts already selected for one state, reducing the local action space to 16, and directly measures every downstream action in that local space.

SHAPE. SHAPE also treats routed expert combinations as coalitions, but its stated methodology does not assume a complete utility function and performs no additional forward evaluations beyond ordinary calibration inference. It uses observed coalition frequency as utility and estimates unobserved subcoalition utility from co-occurrence statistics [Zhang, 2026]. Our local game uses the opposite tradeoff: it is much more expensive per state, but each local subcoalition is actually executed and scored through the frozen downstream model.

Counterfactual route analysis. Yoon et al. [2026] compare a native route with sampled equal-compute alternative routes and show that lower-loss alternatives exist, especially on fragile reasoning tokens. That result establishes that router choice and downstream route utility can disagree. Our experiment is not a global alternative-route search: it freezes the natural top-four identities and exhausts the complete subset lattice inside that active coalition, with final-distribution KL rather than realized-token probability as the primary damage measure.

Subset-aware routing. MP-MoE formulates routing as diversity-aware subset selection and uses expert co-occurrence to model correlations during training [Kang et al., 2026]. This reinforces the broader point that expert selection is naturally combinatorial. Our contribution is a post-hoc measurement study of the exact frozen-model local set function, not a replacement routing algorithm.

Collaboration structure. Tang et al. [2026] identify semantically coherent collaborative expert groups from co-activation patterns across layers. Their result strengthens the evidence that expert function is relational rather than purely individual, but the analysis is based on learned representations of activation structure rather than exhaustive downstream interventions on every subcoalition of one routed active set.

Table 3: Closest prior work and the measurement object studied here. "Exhaustive" refers to executing every subset of the naturally active top-k coalition for a fixed state.

Method	Scope	Utility signal	Exhaustive?	Output
Shapley-MoE	Global expert pruning	Sampled model performance	No	Scalar Shapley estimate
SHAPE	Observed routed coalitions	Routing frequency / co-occurrence	No	Scalar Shapley-style score
Misrouting analysis	Sampled alternative routes	Realized-token probability	No	Route comparison
Alloc-MoE	Layer/token activation budget	Sensitivity + router scores	No	Budget allocation
Ours	Natural active top-k coalition	Final-logit KL after downstream replay	Yes, 2^k	Complete local set function

Budget-aware activation. Alloc-MoE explicitly reduces expert activations under a constrained inference budget, allocating capacity across layers and tokens using sensitivity profiles and routing scores [Liu et al., 2026]. It therefore addresses a deployment problem close to ours. The distinction is the decision signal: our study directly measures the complete within-active-set downstream damage surface and asks which same-cardinality subset is optimal after routing.

The resulting boundary is precise: coalition awareness, collaboration structure, and budget-aware activation are prior art; exhaustive state-conditioned measurement of the complete naturally active top-k execution game is the object studied here.

8 Discussion

8.1 Why conditioning on the natural coalition matters

Exact Shapley evaluation over all n layer experts is exponentially expensive; Shapley-MoE is motivated in part by precisely this obstacle [Huang et al., 2025]. Sparse routing provides a different decomposition of the problem. The router first performs a global reduction from n experts to a small active set k. Conditional on that set, an exact local game requires only 2^k interventions. For gpt-oss-120b this changes the combinatorial table from an impossible full-layer enumeration to 16 local actions. The downstream transformer replay remains computationally expensive, but the coalition enumeration itself is complete.

This does not imply that the router's active set contains the globally best alternative experts. We measure a conditional execution game, not the global routing game. The distinction is a strength rather than a loophole: it isolates the adaptive execution question from the separate problem of changing router identities.

8.2 Why scalar attribution and subset choice differ

A scalar expert score answers an attribution question. Equation (5) answers an action-selection question. They coincide under strong structural assumptions, such as a modular damage function and separable cost. They need not coincide under interaction. Proposition 4 demonstrates that this remains true even when the scalar attribution is the exact Shapley value of a complete game.

This gives a useful interpretation of the empirical failure of singleton damage. The problem is not merely that the wrong scalar feature was chosen; the exact decision object contains interaction placement that no independent scalar score can generally preserve.

8.3 The appropriate runtime target

The current data suggests that a future adaptive executor should predict either the finite damage table itself or a representation rich enough to reconstruct the relevant interactions, then enumerate Eq. (5) against live scheduler costs. For top-four routing, predicting 15 nontrivial damage values or the equivalent Möbius coefficients is still a small-output problem. A practical implementation should also support calibrated abstention and fall back to full top-four execution when uncertainty is high.

The failed ExtraTrees baseline shows that this is not yet solved. In particular, the results argue against integrating an aggressive truncation policy merely because an offline oracle exists.

9 Limitations

First, the exhaustive experiment is currently demonstrated on one model, gpt-oss-120b. Although we intervene at five layers and use two corpora with independent captures, cross-model replication is needed before claiming that the measured interaction frequencies are universal.

Second, the local game is conditional on the naturally routed top-four experts. We do not test expert identities outside that set, so the experiment does not solve router misallocation or global alternative routing.

Third, nonempty interventions renormalize the original router weights over survivors. This is a well-defined causal intervention, but other runtime semantics—for example, retaining unnormalized mass or retraining a router for variable cardinality—define different games.

Fourth, the full-omission action is explicitly defined as zero MoE correction. Consequently, the highest-order Möbius coefficient and any full-game Shapley attribution depend on that extension. Lower-order proper omission measurements do not require the empty-survivor convention.

Fifth, final-logit KL measures distributional fidelity, not task accuracy or realized-token utility. A subset can occasionally improve one realized token while still moving the full distribution away from the baseline. We therefore restrict the primary claim to fidelity relative to the full top-four model.

Finally, exhaustive downstream replay is an offline measurement procedure, not a low-overhead serving algorithm. Our first held-out predictor fails, so the work establishes the target and the oracle gap rather than a production selector.

10 Conclusion

Sparse routing makes a previously prohibitive cooperative-game question locally tractable. Instead of attempting to value every expert in a 128-expert layer exactly, we condition on the four experts that the router has already selected and exhaust all 16 execution subsets. On gpt-oss-120b, the resulting downstream damage surface is strongly non-additive, non-monotonic in a substantial fraction of local comparisons, and materially different from fixed router-rank truncation. At the same executed expert count, exact subset choice exposes roughly a $3.6\times$ mean-KL opportunity over fixed rank policies.

The mathematical consequence is equally important: higher-order interaction is not an optional interpretation layered onto the data. Möbius inversion makes it the exact residual omitted by a singleton-additive model. Shapley values can allocate those interactions fairly to individual experts, but even exact Shapley rankings do not generally preserve the optimal fixed-cardinality coalition.

Adaptive execution should therefore be formulated as a finite, state-conditioned subset decision rather than as four independent expert-importance tests.

The immediate research problem is no longer whether a better subset exists. The exact local game answers that. The open problem is how to predict enough of that game, cheaply and reliably, to realize the oracle opportunity without sacrificing fidelity.

A Formal Verification in Lean

The repository contains `formal/ExactLocalCoalition.lean`. It is checked in CI with Lean 4.32.2. The formal file verifies the following finite four-player algebraic statements:

1. pair, triple, and four-player Boolean-lattice Möbius reconstruction identities;
2. the identity that full-coalition first-order reconstruction error equals the sum of pair, triple, and four-way interaction coefficients;
3. the four-player Shapley–Möbius identity for one player, after multiplying both sides by 12 to eliminate rational denominators;
4. the numerical Shapley values in the counterexample of Proposition 4; and
5. the strict inequalities proving that the top-two Shapley ranking is not the best two-player coalition in that counterexample.

The universal coefficient identities are checked over arbitrary integer-valued symbols. They are integer-linear polynomial identities, so the same algebraic equalities hold for the real-valued damage and utility tables used in the paper. The empirical percentages and KL measurements are not theorem statements and are not represented as formally proved facts.

B Reproducibility Map

The core implementation is versioned in the repository:

- `tools/test_gptoss_subset_utility.py`: exhaustive subset intervention and downstream replay;
- `tools/analyze_gptoss_subset_utility.py`: Möbius transform, reconstruction, approximation errors, fixed-cardinality oracle, fidelity sweep, and runtime objective sweep;
- `tools/predict_gptoss_subset_utility.py`: held-out cheap-feature prediction baseline;
- `tests/test_gptoss_subset_utility.py`: renormalization and numerical Möbius unit tests;
- `reports/gptoss-subset-utility.md`: aggregate experimental report;
- `formal/ExactLocalCoalition.lean`: formal algebraic checks.

The large per-state replay outputs and model checkpoint shards are intentionally not committed. Aggregate results and reproduction code are versioned.

References

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